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WoT-Server: Combining the Web-Of-Things and SensorThings standards to build an IoT platform

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Summary

This document presents WoT-Server, a JavaScript IoT server that combines the abstraction of the Web-Of-Things Standard and the data model of the SensorThings Standard to provide a simple interface for storing and requesting data in timeseries format. WoT-Server provides simplified access to the different instances required for organised storing of IoT timeseries data via easily accessible HTTP endpoints, which can also be adapted to be accessed with different protocols such as CoAP and MQTT. This paper describes the concepts and the techniques used in the development of this tool. It also shows how to use it in a water management/irrigation application, presenting the original use case developed in the ACQUAOUNT project (Acquaount, 2024), a project funded by the European Union with the objective of providing tools for efficient use of water.

Statement Of Need

WoT-Server was created as a centralised data management solution for the ACQUAOUNT Project. One of the objectives of the ACQUAOUNT Project is providing smart irrigation recommendations to farmers using a water balance model. This model takes as input timeseries data of different properties of the specified field, such as temperature, soil moisture and wind speed, and calculates the best date and amount of irrigation. WoT-Server was developed as a standardized API to allow easy access to faming/water management data for further model implementation and to simplify the upload, storage and retrieval procedures of measurements from sensors in the field. WoT-Server is capable of both receiving data via an HTTP endpoint or fetching the data itself from other API services, a feature useful for centralising data from multiple platforms.
All AI-based applications require data, and most if not all IoT applications obtain this data from sensors in timeseries format. The aim of WoT-Server is to provide a platform to easily manage this type of data with minimal setup and configuration required, letting users focus on the tasks specific to their use case. With WoT-Server, all developers creating IoT applications can have a platform where they can store their data up and running in minutes, which they can use to create AI models.

Design And Implementation

The platform, according to the requirements of the project, must consist of an API implementation based on WoT descriptions that controls the heterogeneity of the different sources of data (such as different types of in-situ sensors or different weather stations) by providing a common data exchange model. This solution attacks one of the previously mentioned problems of IoT, the platform fragmentation, and defines how a sensor should communicate its data to the platform.

To provide all the required functionalities of the project in a clean, scalable, and efficient manner, the platform is divided into multiple services, all of them virtualized and deployed using Docker. A basic version of the different services and their connections can be seen in Figure 1.

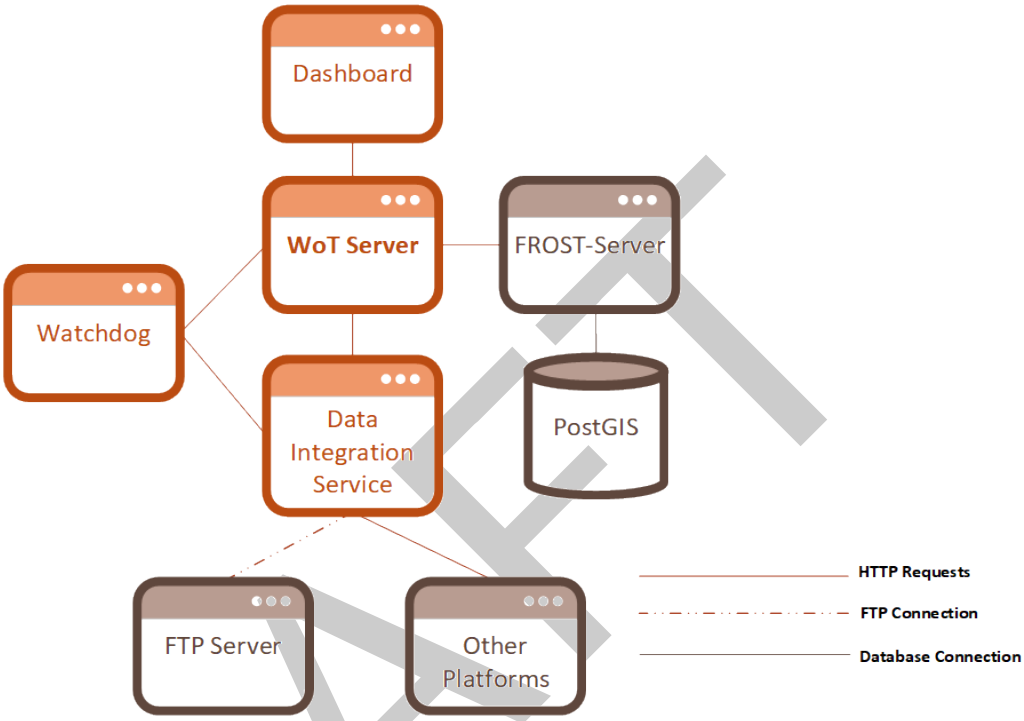

Figure 1: Diagram of the WoT platform and all related services.

The main service is the WoT server, developed using node-wot (Eclipse Thingweb, 2024). This service provides all the endpoints and the controllers that handle the requests sent by the sensors and the gateways. To provide persistent storage of the data, a PostGIS database is used. However, to simplify the development process and to maintain consistency with the standards, FROST-Server, a SensorThings compliant server, is used to access the database (Fraunhofer Institute, 2024). Then, a different service is used to integrate data from external platforms into the WoT platform, either by directly requesting it to the external platform or by fetching it from an FTP server where it has been uploaded. In addition, the watchdog service is used to control if the server and the data integration service are online and haven't crashed and gone offline. If this happens, the watchdog sends an email to the server administrator to restart the service as soon as possible to avoid missing out on data being sent from the sensors. Finally, the monitoring platform provides a simple way to check if data is correctly being uploaded to the server.

Inside the WoT server, it was decided that the best course of action would be to define Things in the platform as the closest possible match to a physical location. Each physical location can be interacted with by following the WoT protocol and interaction affordances described in the README.md file. These interaction affordances act as a bridge between the user and the data stored in the SensorThings database. Since the WoT model and the SensorThings model share the Thing as a common instance, they can collaborate to create a model that is complete and detailed but abstract and simple to interact with at the same time, as seen in Figure 2.

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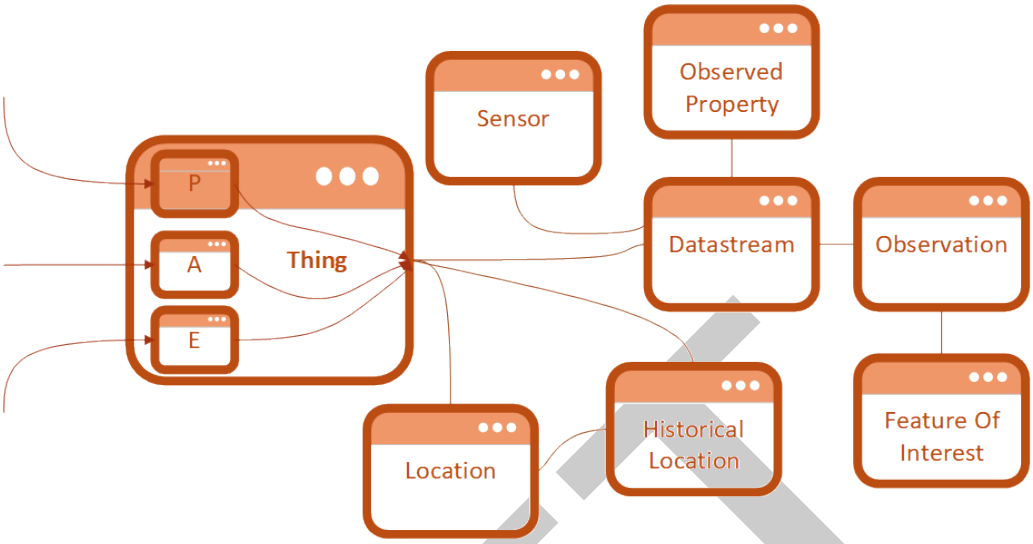

Figure 2: Diagram of WoT being used as the interface to access the SensorThings data model.

Use Case

WoT-Server has been developed as one of the core tools of the ACQUAOUNT Project, a project funded by the European Union via the PRIMA program and supported by Horizon 2020, with the objective of developing a variety of innovative tools focused on making the use of water at farm level and basin level as efficient as possible. These technologies are developed and tested in 4 areas of the Mediterranean area (Sardinia, Italy; Bekaa Valley, Lebanon; Al-Jifārah plain, Tunisia; Central Jordan Valley, Jordan), seen in the map in Figure 3.

Figure 3: Map of the Pilot Locations of the ACQUAOUNT Project.

The ACQUAOUNT Project has many lines of research and development, such as sensor installation, LoRaWAN networks, or basin modelling. WoT-Server acts as the link between the sensors and platforms and the models, storing data and providing functionalities in a standardized way. WoT-Server stores the data received from the data sources and provides it to services that calculate recommendations based on this data. Once the services finish the calculations, the results are also stored in WoT-Server to be consumed by the dashboard and frontend services to be displayed to the final users in a graphical and understandable way.

Acknowledgements

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